PROVE

prototype vehicles laboratory

High Voltage Battery Pack
Critical Design Review

Spring 2026



Project Introduction

Project Summary

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"The car currently does not have a permanent structure to hold the high voltage batteries. Our project seeks to create a structure that can hold the batteries in their designated spot in the car. The structure is a crucial part of getting the car prepared for full driving tests and further development"

Battery Pack Goals

Primary Objectives

- Store all batteries in the appropriate spot
  - 5 batteries per module
  - 8 modules in front box, 6 modules in rear box
- Provide electrical mounts, routing and accommodation

Secondary Objectives

- Allow people/teams who need to access the batteries and other components to do so easily
  - Structure should be fully disassembled by two people with basic hand tools
- Protect all relevant components from sources of damage/failure
  - Batteries, electrical components
  - Car dynamics (breaking, turning), thermals, vibration, expansion

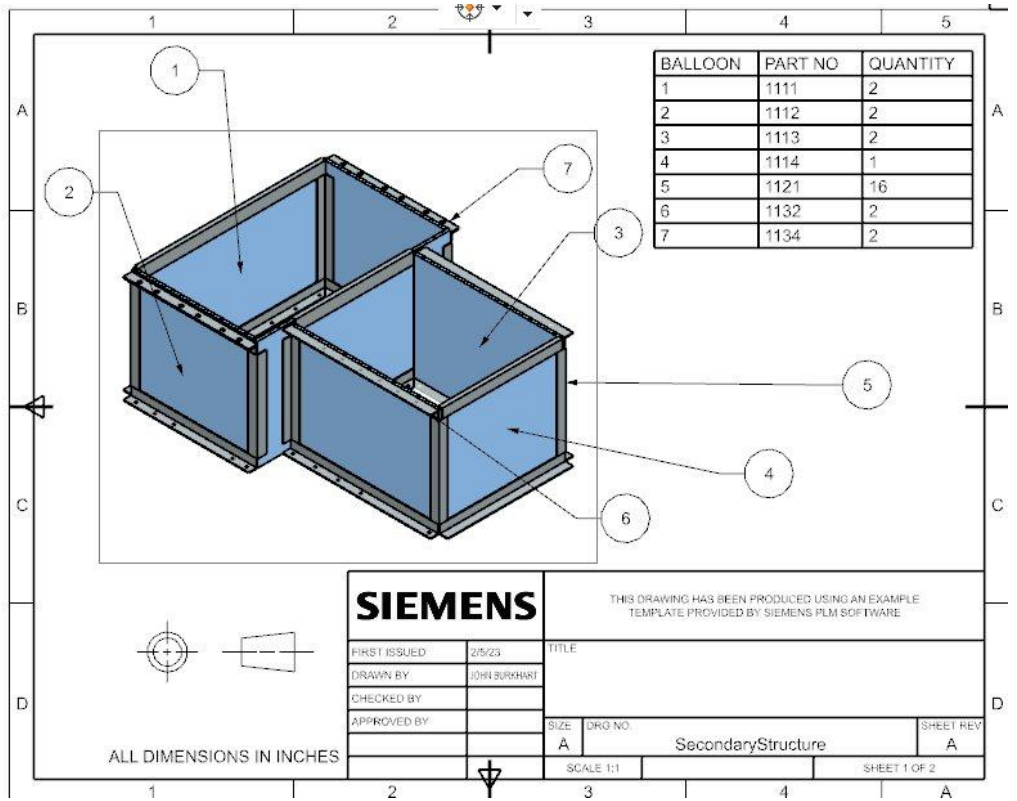
Interfacing

Electrical

- BMS
  - Wired to each battery separately
- Contactor
  - Connects high voltage signal
  - 2 gauge wire (6.54mm dia)
- Bus bars

Mechanical

- Current battery box (Battery Structure CDR, 2022)
  - MDF + carbon fiber coating
  - Not perfectly square
  - Designed for 1.5g crash



Requirements

Structural

- Structure must be able to move <1/16” in each direction once secured
  - Maximum 1.5mm clearance
  - Test fitting/inspection
- Each module and battery must be allowed a clearance fit within each other such that they can be removed easily

Electrical Interfacing

- All battery wiring + connection points must be approved by HV
- All batteries must be electrically isolated

Requirements

Performance
- All batteries must be able to be operated on individually

Safety
- Batteries should maintain temperature under 130°F
- Full structure should be disassembled in less than ten minutes with minimal tools

Organizational
- Print modules for electrical testing by end of Spring 2026
- Finish full pack assembly by beginning of Fall 2026
- Keep full pack cost under $1,000

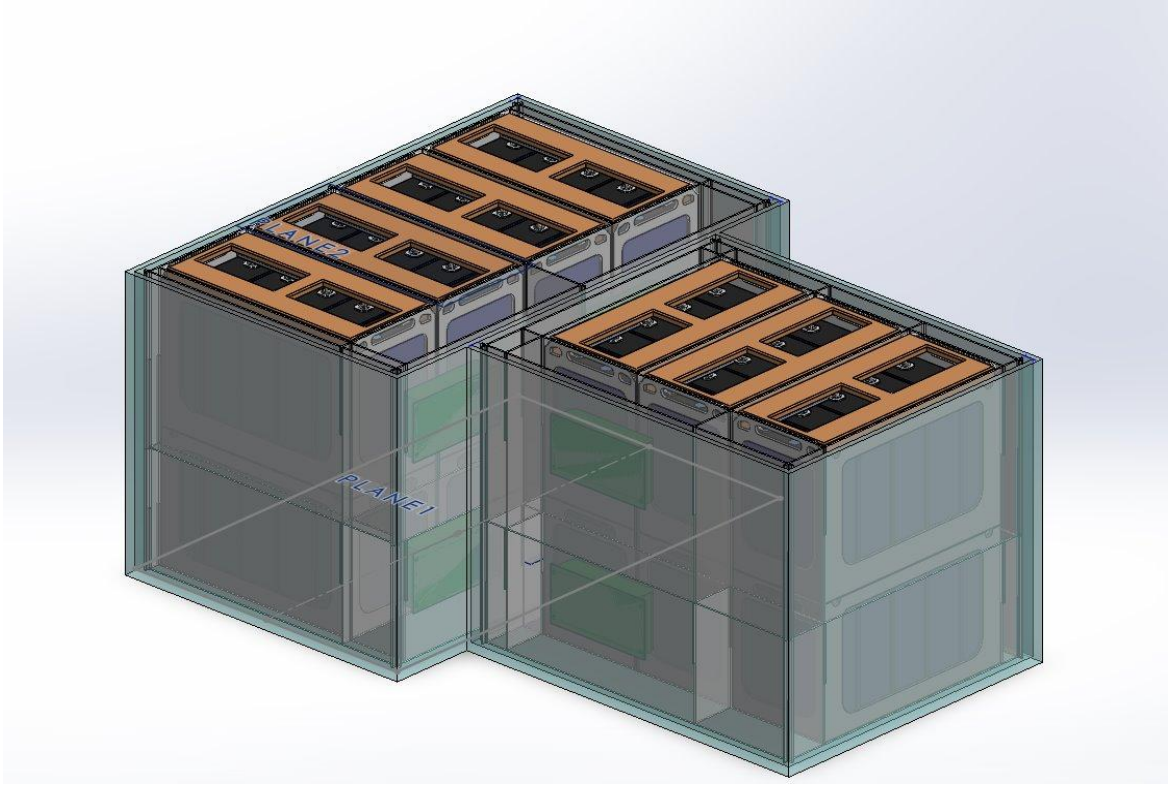


Design Overview

Full Structure

Full Structure

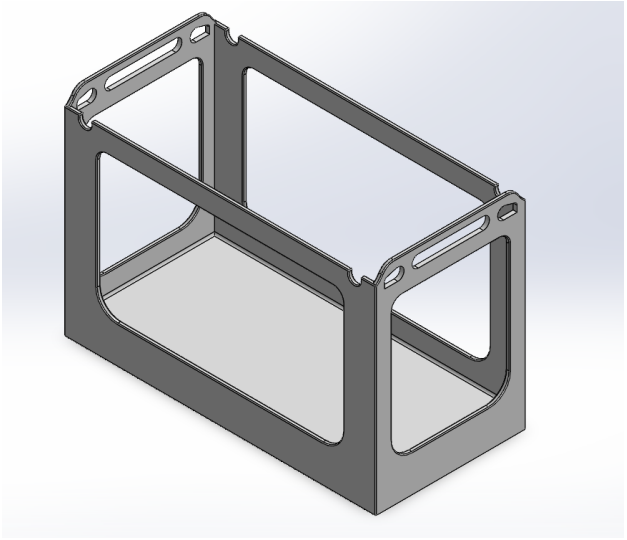
- Total of 70 EVE LIFEPO4 314Ah v2 batteries
- Movement restricted by battery box mounted to frame of vehicle

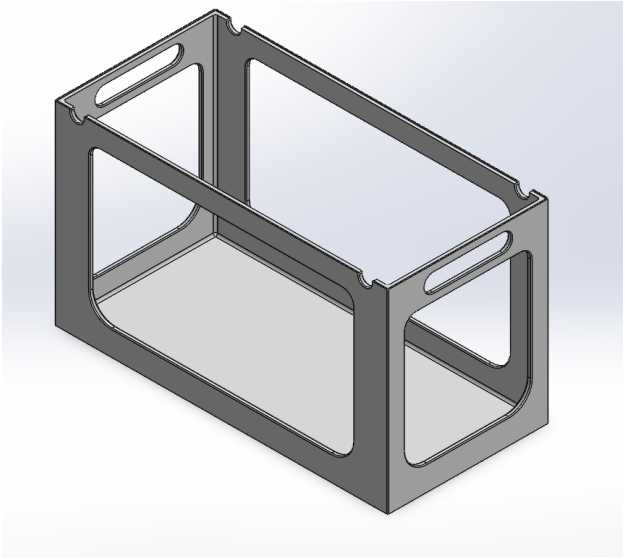


Battery Module

Battery Modules

- Each house 5 EVE LIFEPO4 314Ah v2 batteries
- Immediately responsible for restricting movement of batteries

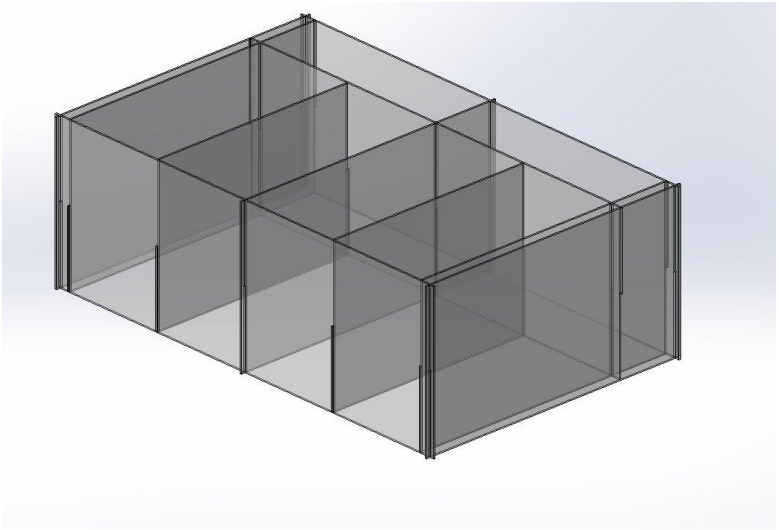


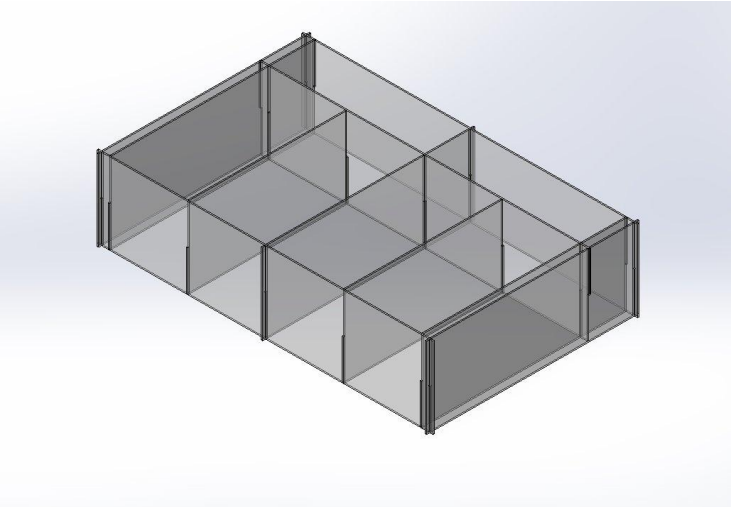


Front Waffle

Front Waffle

- Must constrains Modules horizontally
- Houses 8 modules and mounts 6 electrical components

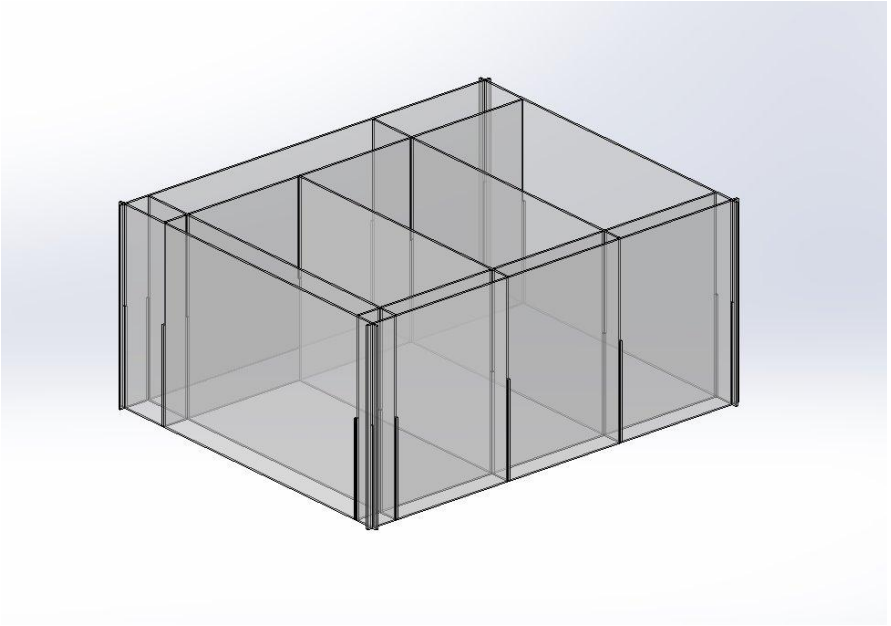


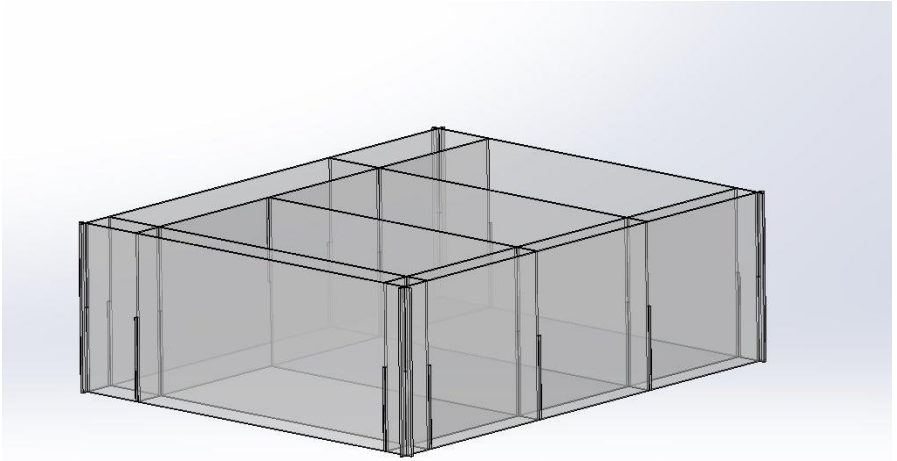


Rear Waffle

Rear Waffle

- Must constrains Modules horizontally
- Houses 6 modules and mounts 4 electrical components
- Designed with removability in mind for both waffle and modules

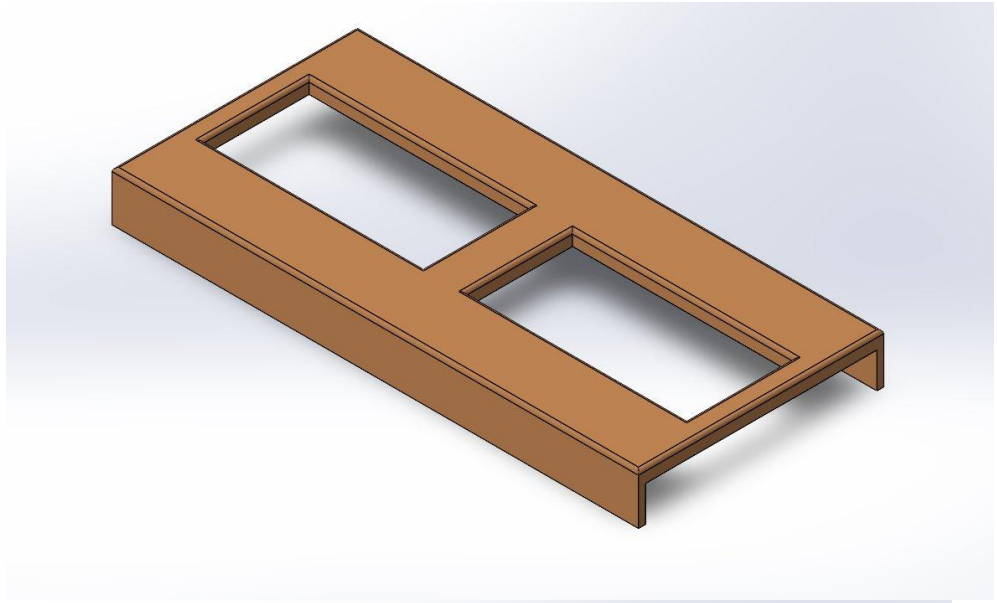


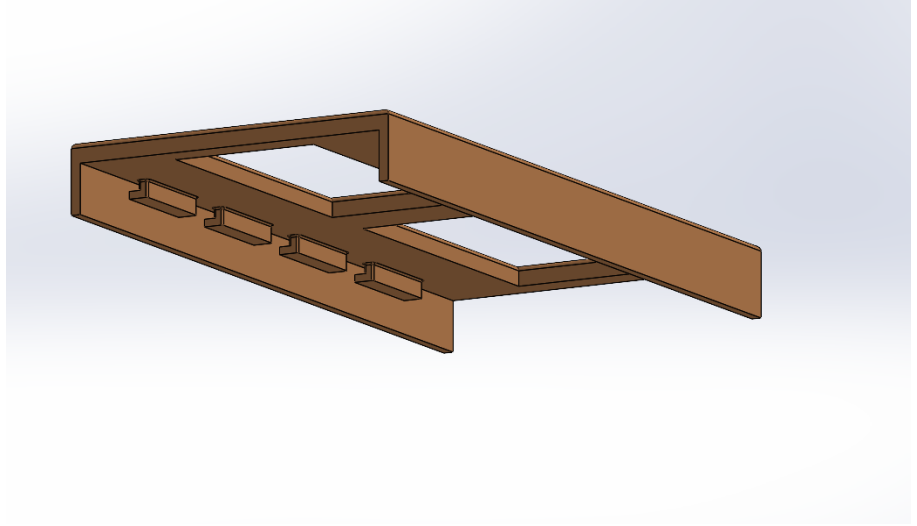


Wire Manager

Wire Manager

- Constraints batteries in the Z direction
- Manages wires going to batteries in an organized fashion
  - Prevents BMS wire from touching battery terminals





Electrical Components

Electrical Components

- Mounted to waffle
- 6 BMS (4 in front box, 2 in rear box)
  - Wired to each battery separately
- 6 Contactor (4 in front box, 2 in rear box)
  - Connects high voltage signal
  - 2 gauge wire (6.54mm dia)

Design Analysis

Major Design Considerations

Thermal
- Battery heat dissipation
- Maintaining integrity across thermal spectrum

Structural
- Vehicle dynamics

Design for Manufacturing/Assembly
- Waffle structure
- Quick removal procedure

Subsystem Interfacing
- Electrical mounts



Thermal

Battery Heat dissipation

Battery Max Temp: 140°F

- Foam max temp: 190°F (FOS 1.35)
- ABS T_g: 275°F (FOS 1.96)

Structural

Vehicle Dynamics Assumptions

- Acceleration of 3g in all directions
  - Lateral, Longitudinal, Vertical
  - External battery box rated for 1.5g

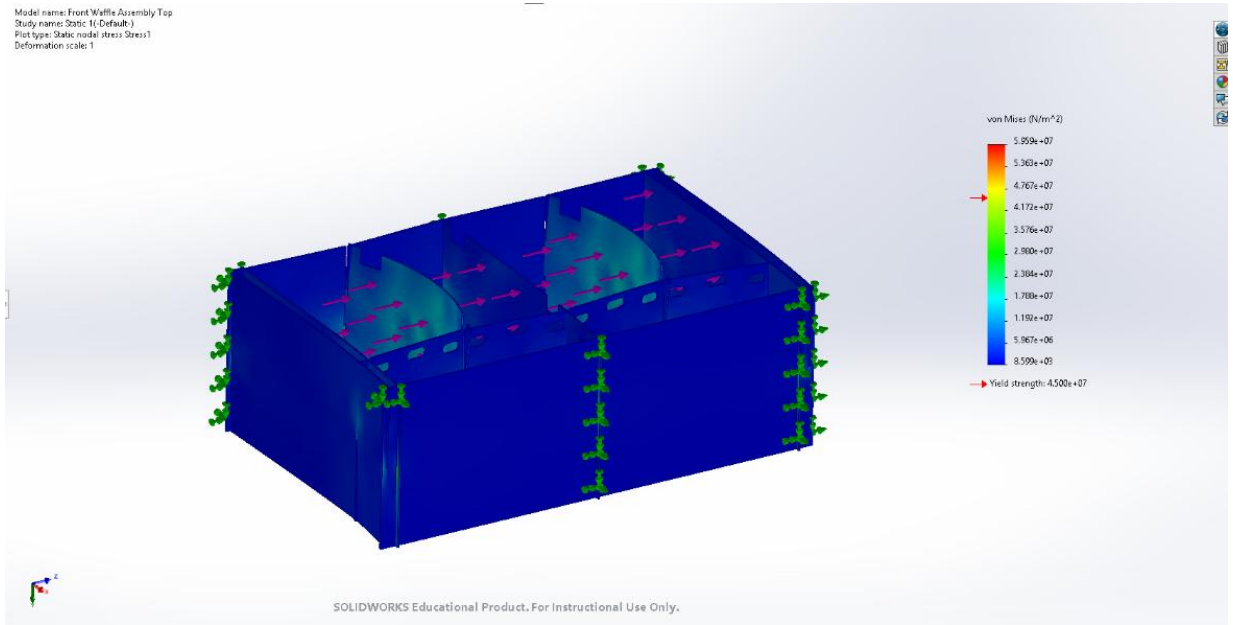
- External Connections are rigid

- Polycarbonate sheets have tensile strength of 54Mpa
  - (Grainger Datasheet)

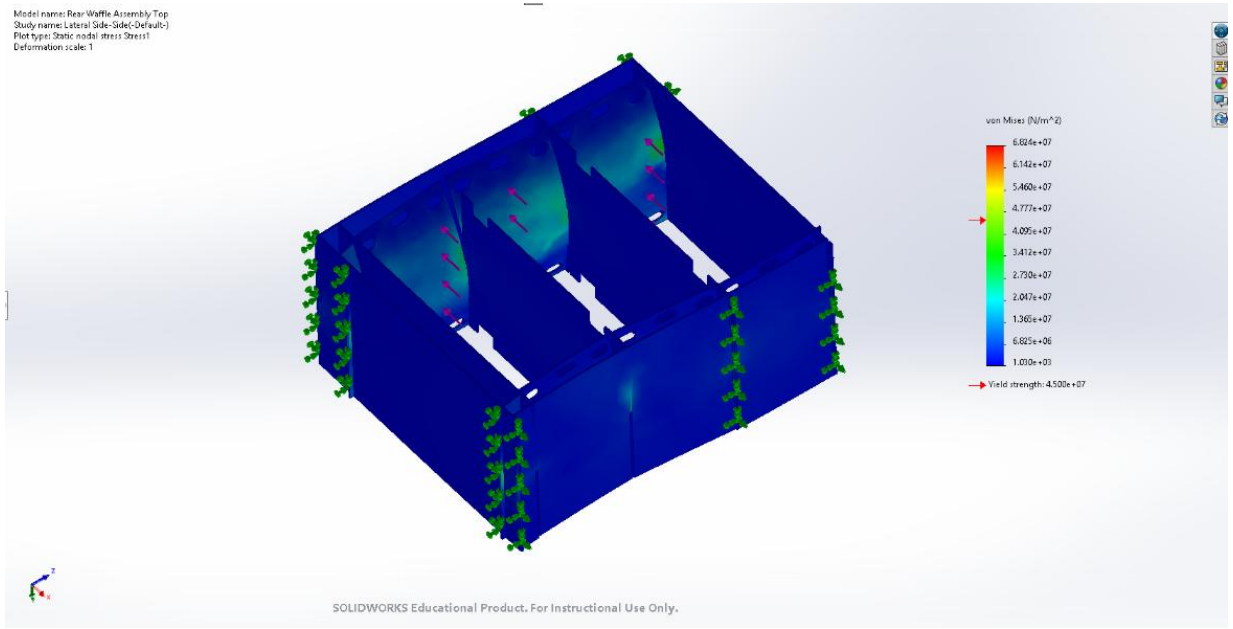
Structural

Lateral Acceleration

Front Structure



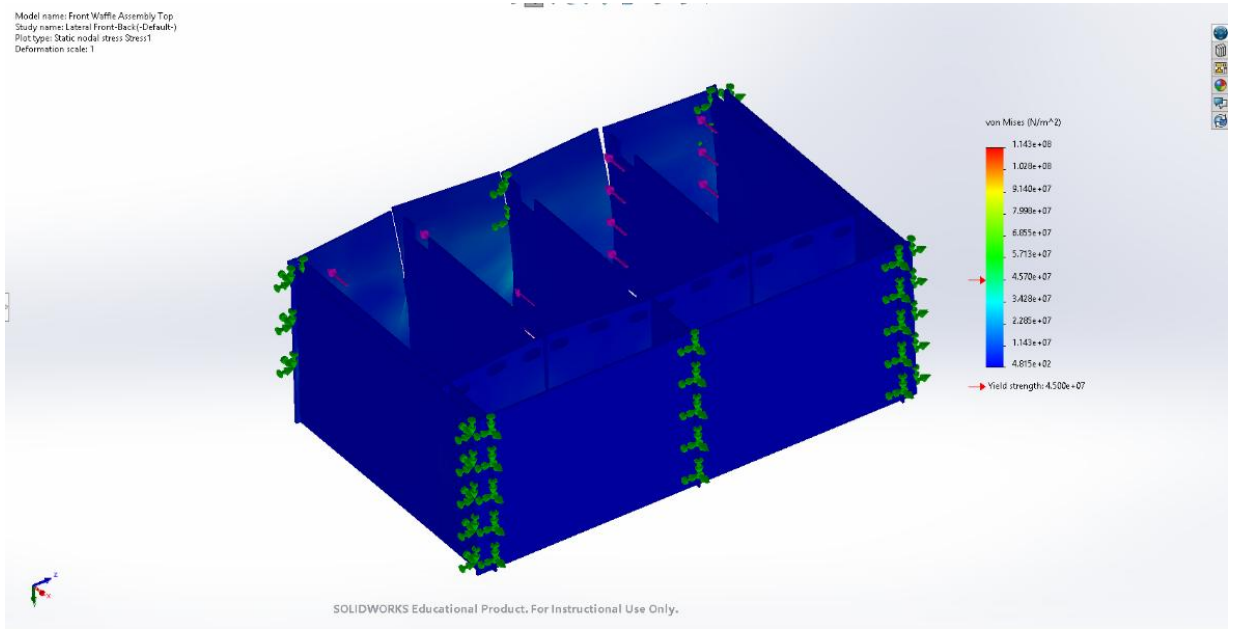
Rear Structure



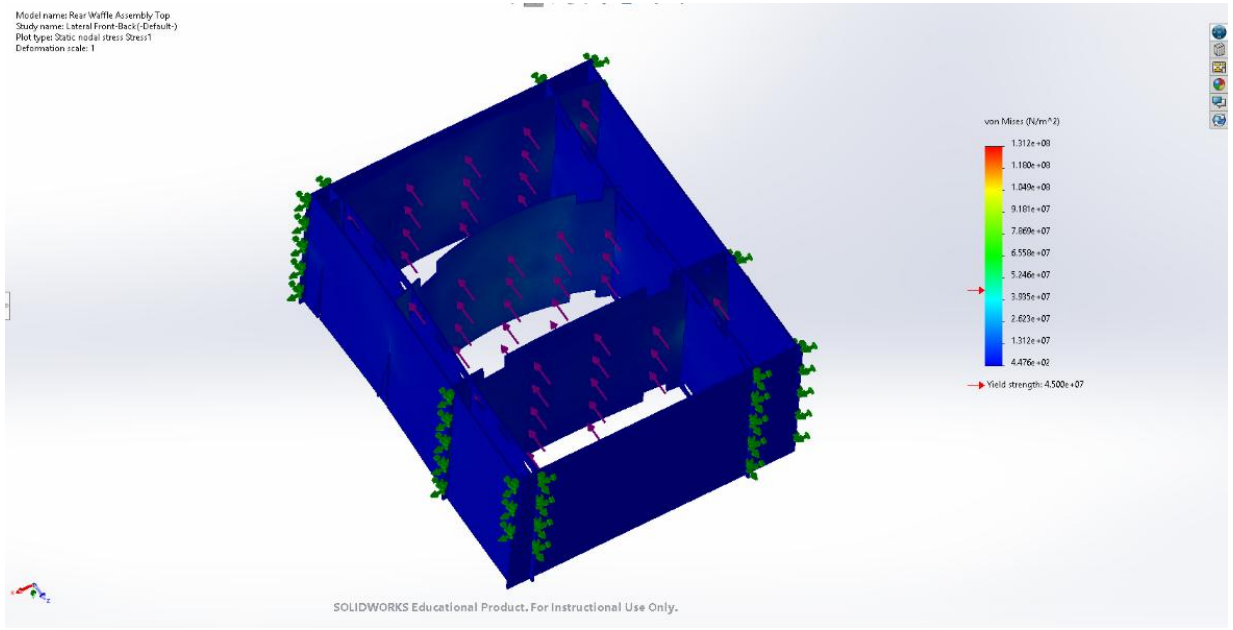
Structural

Longitudinal Acceleration

Front Structure



Rear Structure

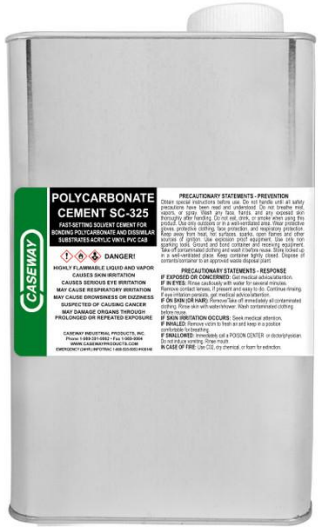


Design for Manufacturing

Waffle Structure Design Paradigm

- Each wall is made from 1/8” polycarbonate
- Walls slot together

- Bonded using Caseway SC-325 solvent
  - Chemically fuses parts into one seamless piece
  - Flammable before fully drying

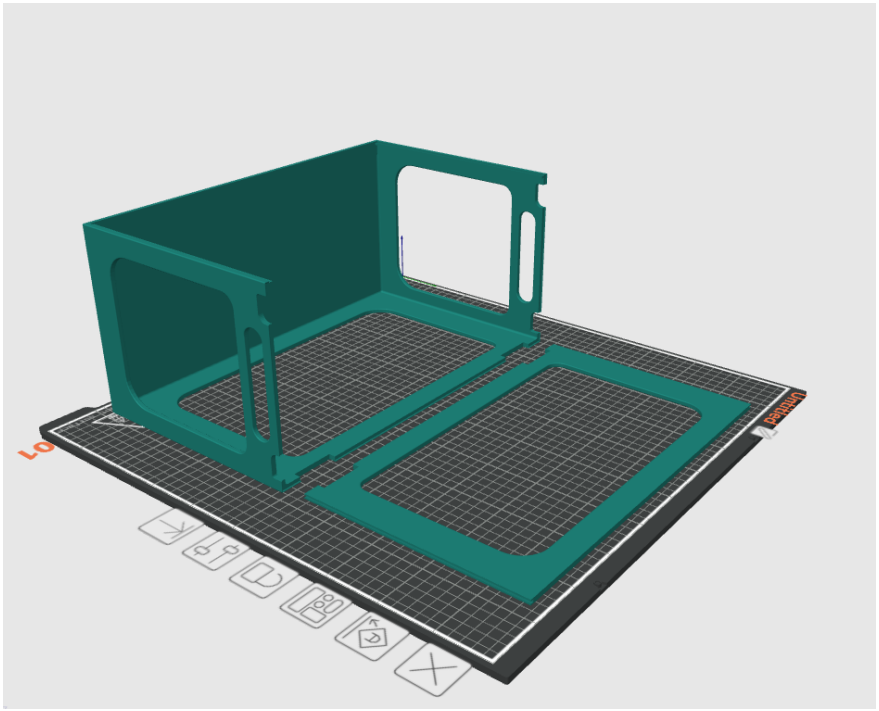


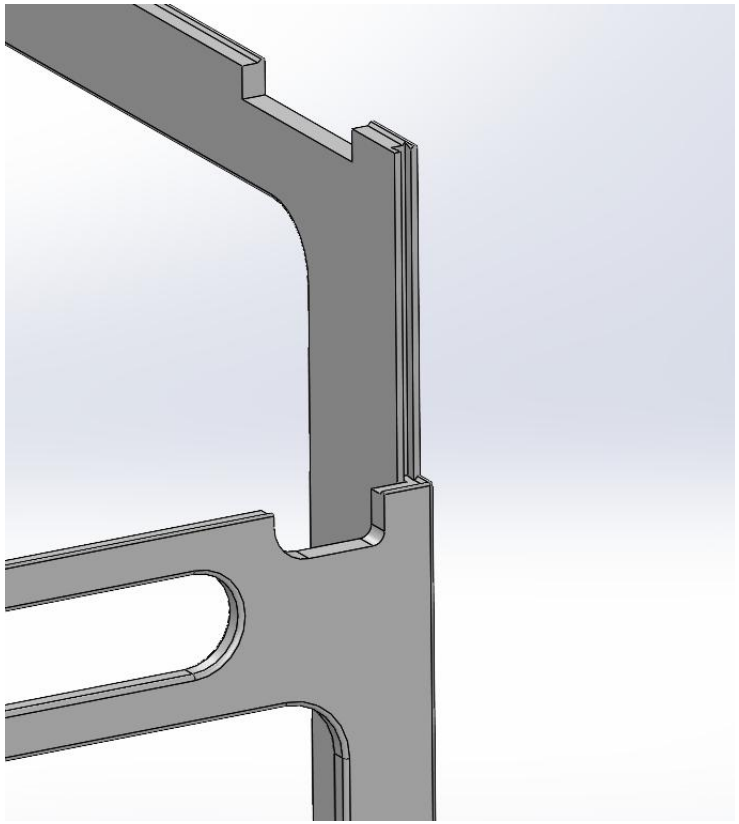
Design for Manufacturing

Module Printing

- Print modules in 2 sections

- Dovetail slot for joining sections
- Loctite HY 4090 adhesive for joining

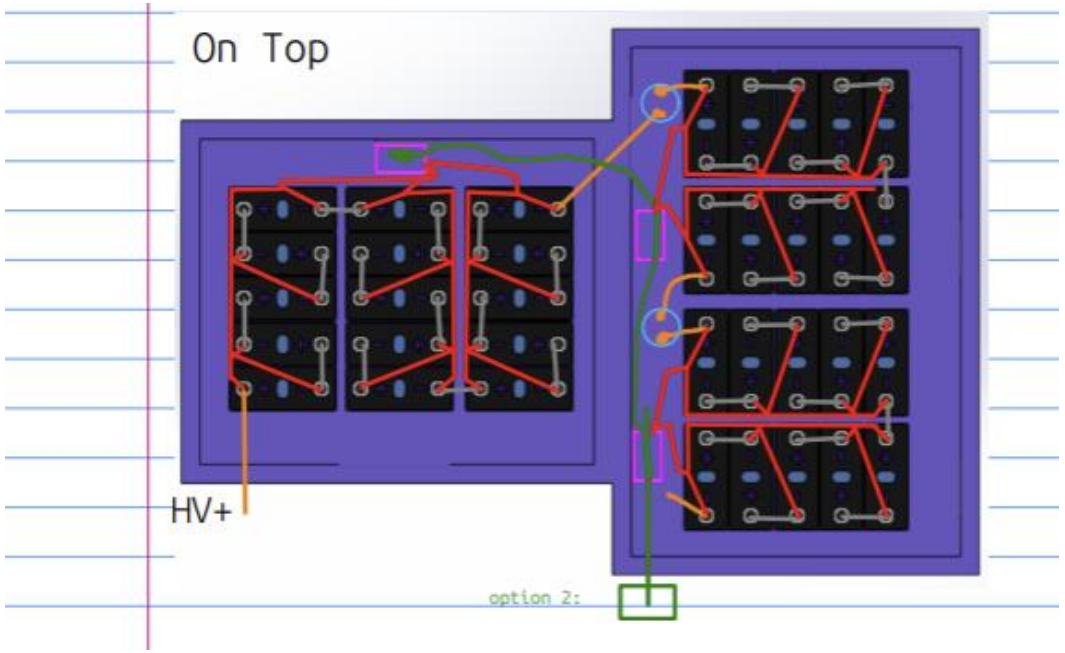


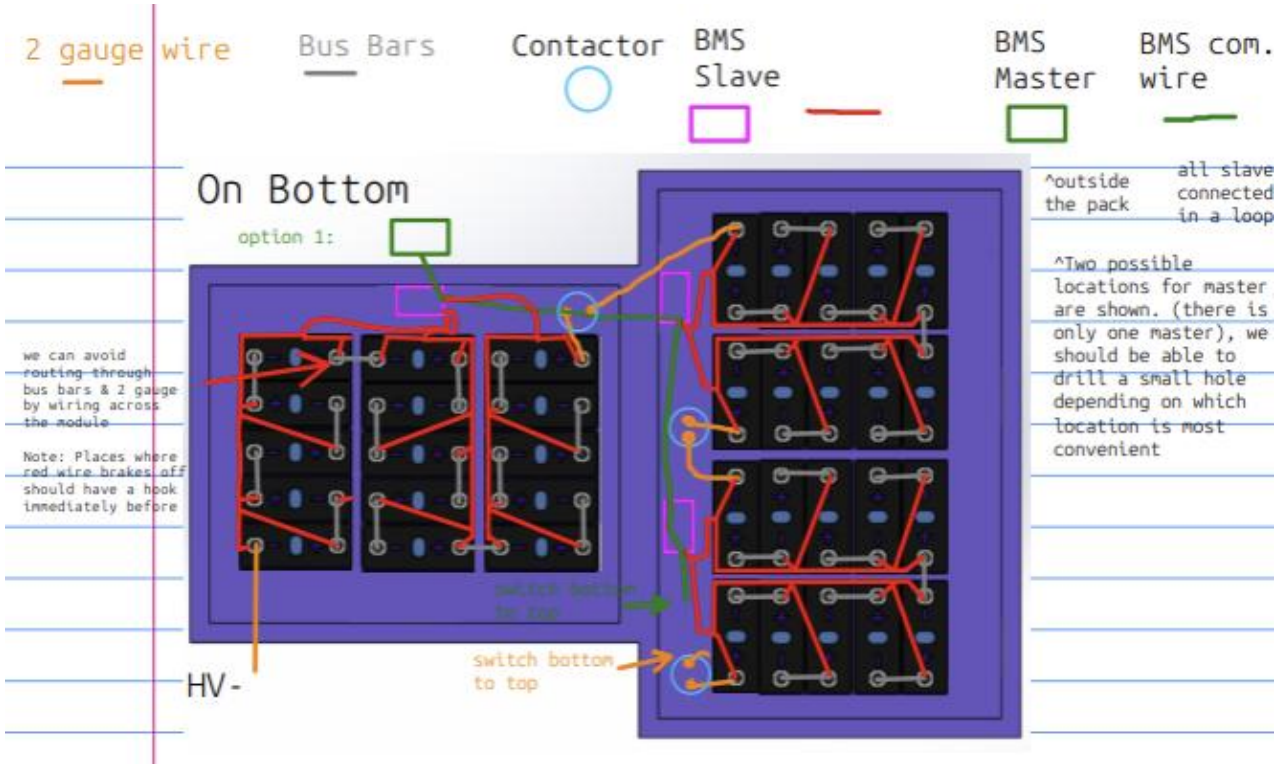


Design for Manufacturing

Electrical Mounting

- Front structure (per level)
  - 2 contactors, 2 BMS
- Rear structure (per level)
  - 1 contactor, 1 BMS





- Holes in waffle structure allow for wire management
- Placements in full model match general diagram locations

Design for Manufacturing

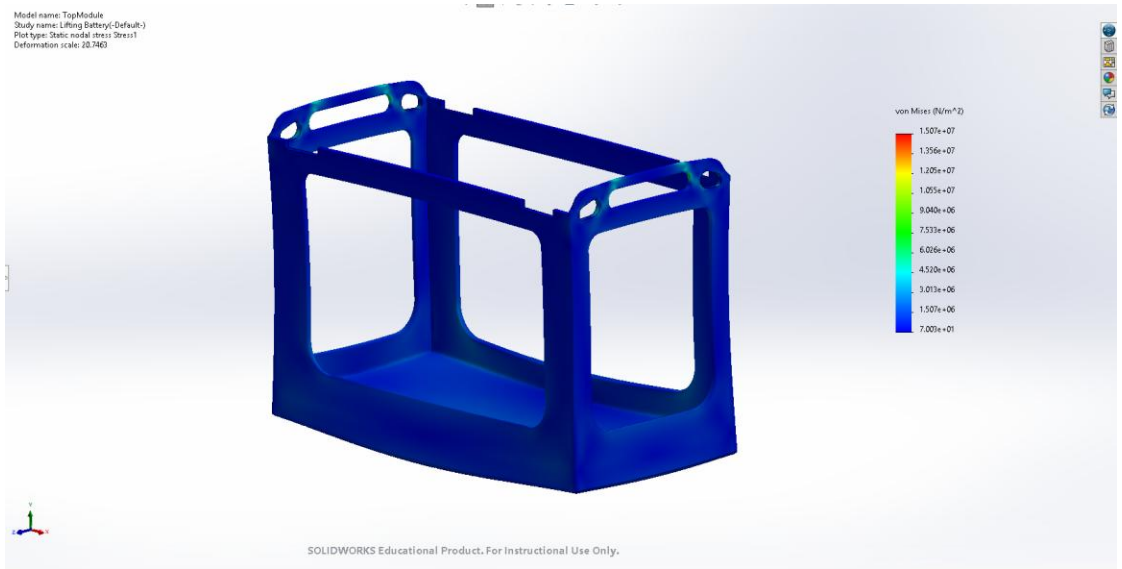
Module removal

- Maintain strength to remove batteries during removal

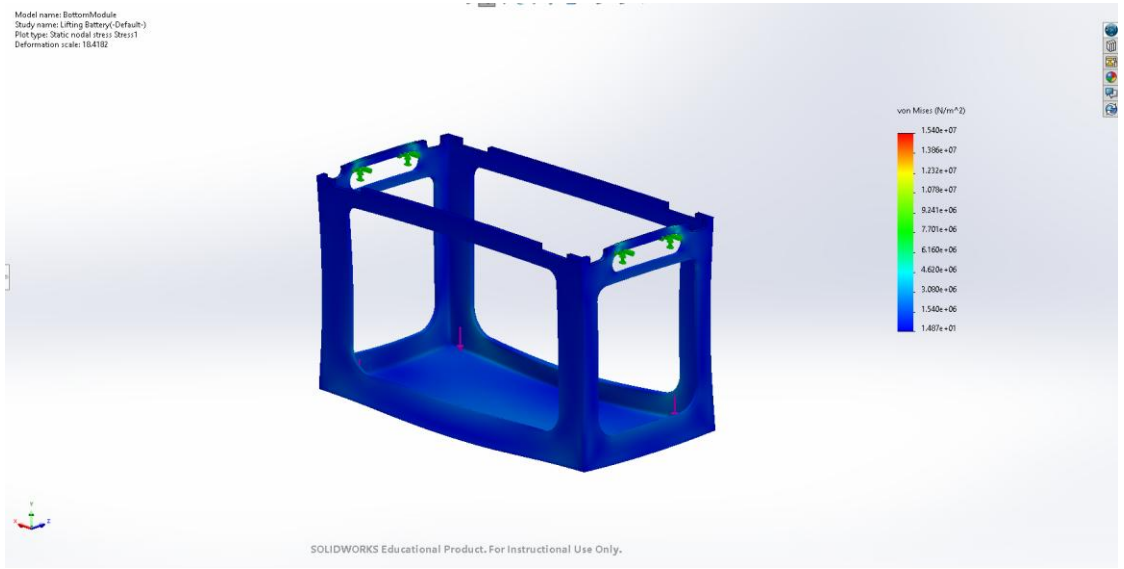
Assume:

- Battery mass: 5.4kg
- Upward acceleration is negligible

Top Module



Bottom Module

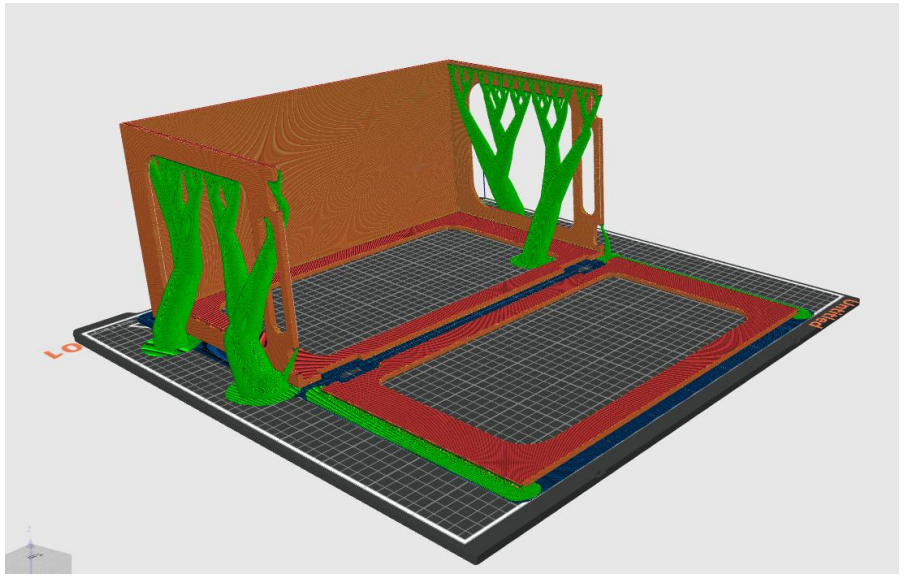


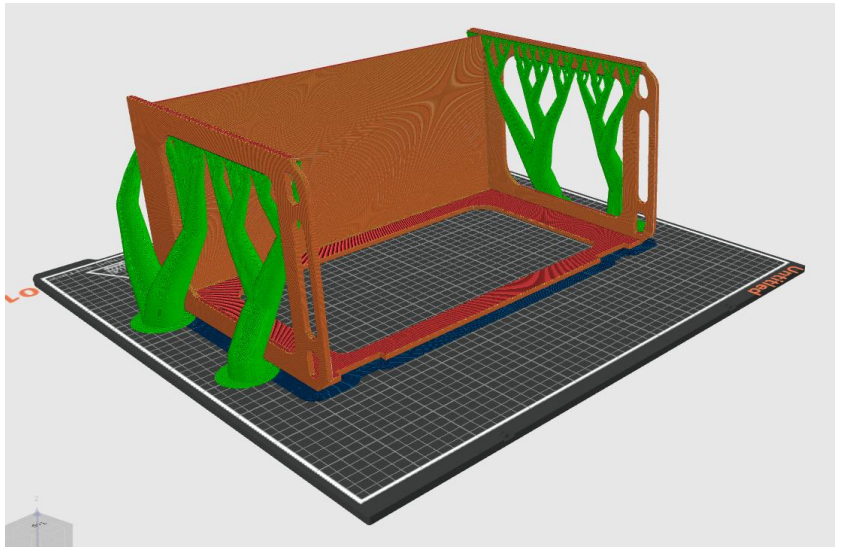
Manufacturing and Assembly

Modules

3D printed in two parts using PROVE printer
- Bottom module can be one print
- Top module must be two prints (larger than bottom)

Parts slotted into each other
- Dovetail slot
- 0.5mm clearance fit (room for epoxy)





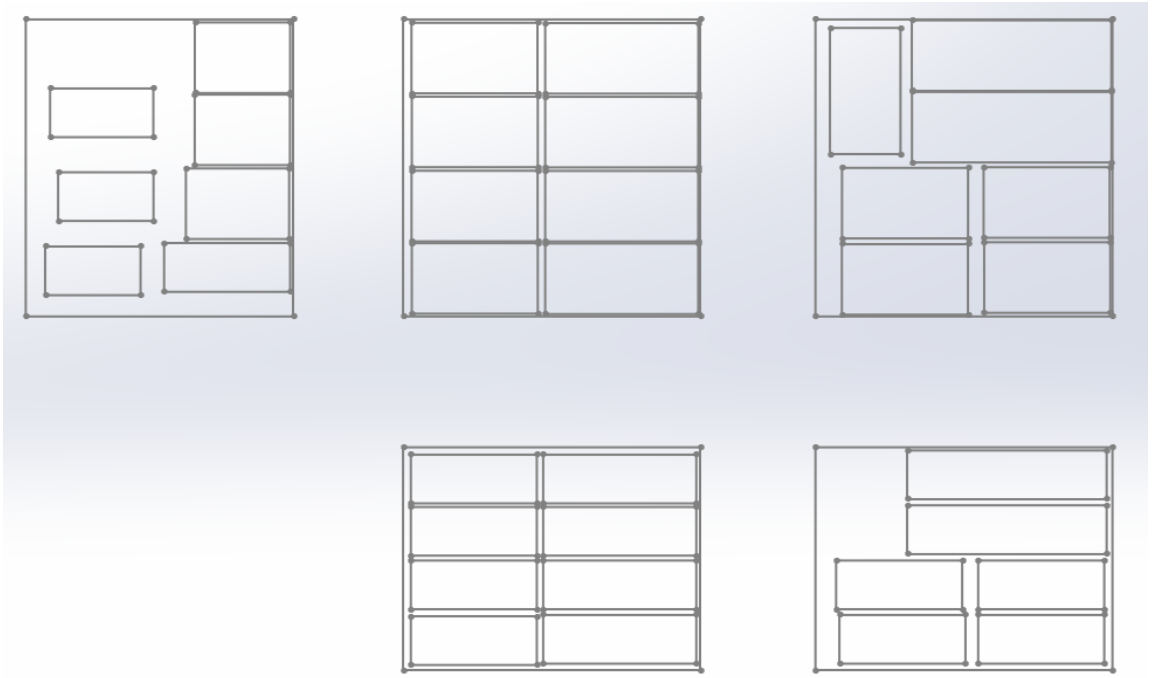
Waffles

Manufacturing

- Water Jetted out of 1/8" polycarbonate sheets
- Slots together and bonded using Caseway solvent cement

Document: “Polycarbonate Fits” details how polycarbonate should be cut from sheets

(Full document includes dimensions)



Wire Manager

Manufacturing

- 3D printed in PLA to be electrically insulating and does not need to be as strong as Modules

Testing

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Thank You

